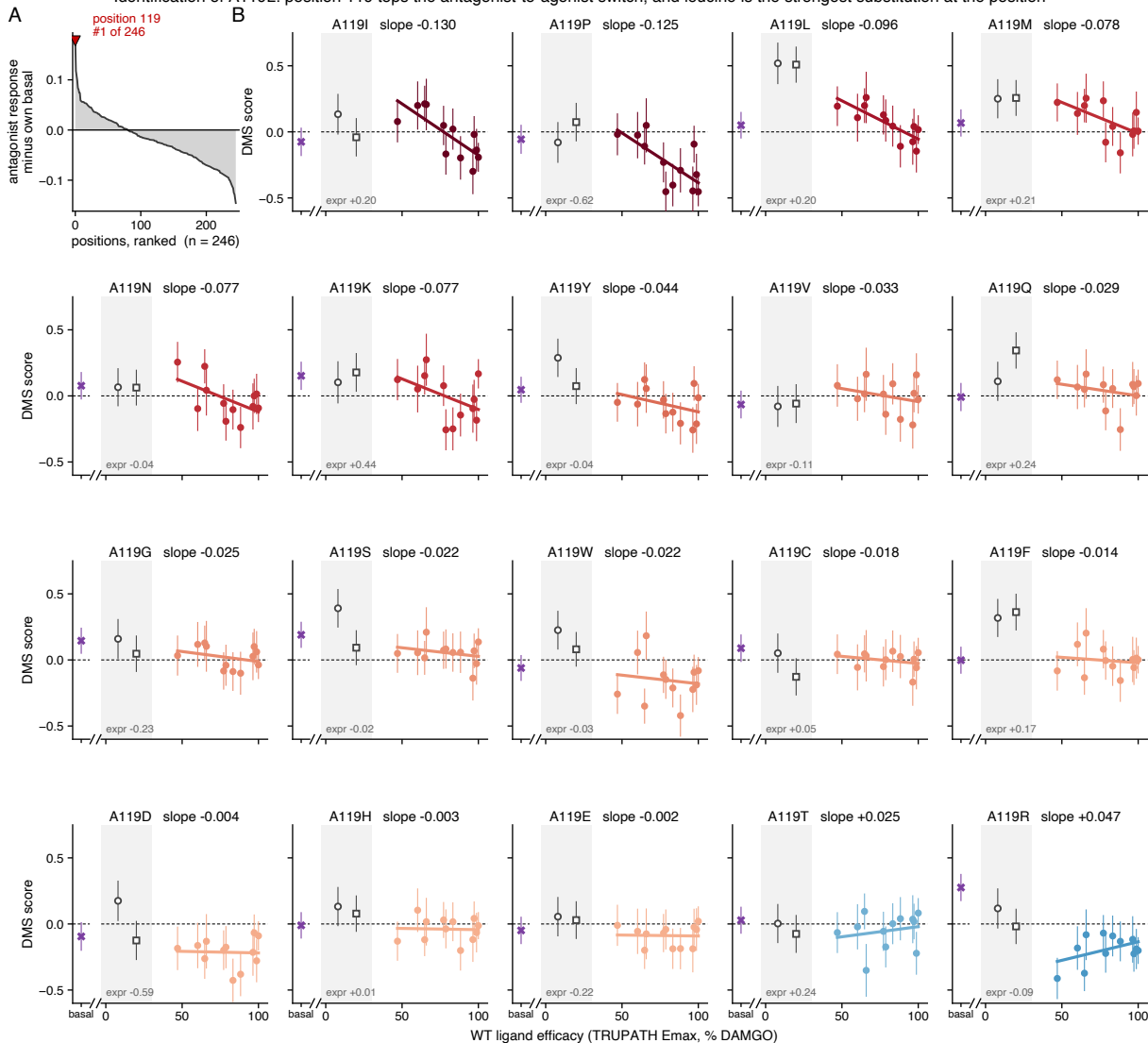


Identification of A119L: position 119 tops the antagonist-to-agonist switch, and leucine is the strongest substitution at the position



A positions ranked by mean antagonist response minus that variant's own basal. Naloxone and naltrexone are required to agree to within 0.05 and a position needs ≥ 5 surviving substitutions to enter the ranking; red triangle = position 119.

B one panel per substitution at position 119, ordered by efficacy slope (most negative first). Coloured points = the agonist series placed by WT ligand efficacy, solid line = fit to the agonists only. Open circle = naloxone, open square = naltrexone, drawn apart so the two antagonists can be compared. Purple X on the separate left segment of the x axis = basal, no ligand. Antagonist points above it are ligand-driven; level with it, constitutive.